NeoSignal

AI Intelligence Daily · Multi-Source · Cross-Verified

21 July 2026

200 verified · 366 raw · 5 sources

200	2	198	0
TOTAL	VERIFIED	CONFIRMED	EMERGING

Pipeline

Scrape HackerNews (Show/Top/New), Reddit (r/artificial, r/MachineLearning, r/singularity, r/LocalLLaMA), TechCrunch AI, VentureBeat AI, MIT Tech Review, The Verge AI, Wired AI, ArXiv CS.AI

Filter 40+ AI keyword patterns — LLMs, alignment, safety, research, tooling

Dedup Title-similarity (45% threshold) merges cross-source variants

Score Authenticity = 0.5 base + 0.3 per extra source + diversity + HN score bonuses

Rank Sorted by authenticity. Below 0.25 threshold are dropped.

Authenticity Tiers

VERIFIED	**0.80 - 1.00**	3+ independent sources including media
CONFIRMED	**0.50 - 0.79**	2 sources or 1 high-quality media
EMERGING	**0.25 - 0.49**	Single-source; passes AI keyword filter

Source Breakdown

Source	Articles	Category
ArXiv CS.AI	180	Media
HackerNews	8	Community
TechCrunch AI	7	Media
The Verge AI	5	Media
MIT Tech Review	2	Media

VERIFIED INTELLIGENCE — 2 stories

VERI

Show HN: I built claw-coder which is the first atonomous local AI agent

HackerNews +1 source 90% HN 3

I built PounceDomains to find undervalued domains on the Namecheap Marketplace before others. It scans domain auctions based on your configurations around the clock, AI-scores each candidate against your buying strategy, enriches matches with authority/backlink/demand data. Free...

https://news.ycombinator.com/item?id=48982875

VERI

Position: Explanation Stability Is a Property of the Model Method Pair, Not the Model

ArXiv CS.AI +1 source 80%

arXiv:2607.16652v1 Announce Type: cross Abstract: This position paper argues that claims about explanation stability are scientifically invalid without cross method validation. Just as statistical significance requires the test statistic to be specified, stability should either...

https://arxiv.org/abs/2607.16652

CONFIRMED SIGNALS — 198 stories

CONF

Show HN: A Pipeline for Making 10-minute AI Movies with Claude Code and Seedance

HackerNews 50% HN 17

I've been working on this Claude Code pipeline for making 10-minute movies, gluing together Seedance (videogen), Nano Banana (image gen), and ElevenLabs (voice gen), and using Claude as the director. My repo contributes a markdown playbook and a worked example which should make...

https://github.com/dawndrain/movie-gen

CONF

Show HN: Sharper – an AI office agent grounded in your knowledge, with citations

HackerNews 50% HN 3

No summary — see link for full article.

https://sharper-ai.co

CONF

Show HN: Relay – a self-hosted LLM gateway with eval-gated routing

HackerNews 50% HN 3

Hey HN! Wanted to show off a thing I built in case it interests anyone. I wanted to build a game you play inside the claude code CLI and back when they launched the channels feature I started investigating ways to use that to do so. I started with a multiplayer comedy battle...

https://github.com/llmrelay/relay

CONF

Show HN: AI World Bakeoff – 9 models × 3 Three.js worlds

HackerNews 50% HN 3

With the releases of Fable 5, gpt 5.6 sol, and Kimi K3 I was curious how these models would perform building vivid 3d worlds, especially a Chinese open-weight one like Kimi. I also threw in some budget models so you can see how drastic the quality differences are.

https://ai-world-bakeoff.pages.dev/

CONF

Show HN: Lagotto Meter – how well do AI agents understand your website?

HackerNews 50% HN 2

Built this to measure the semantic delta between what a site declares and what an AI agent actually reconstructs. Free, no signup. I'm the author — happy to answer questions.

https://lake8.dev/lagotto-meter/

CONF

Show HN: Agent Search Engine – an independent index of 247 AI agents

HackerNews 50% HN 2

No summary — see link for full article.

https://agentsearchengine.app

CONF

Show HN: Intersections – A Daily Word Game

HackerNews 50% HN 2

No summary — see link for full article.

https://www.playintersections.com/

0 CONF

Anthropic's landmark $1.5B copyright settlement is approved

TechCrunch AI 50%

The final approval settles one case, but it doesn't resolve the broader issue of using copyrighted works to train AI models.

https://techcrunch.com/2026/07/20/anthropics-landmark-1-5b-copyright-settlement-is-appr...

1 CONF

Trump's latest AI czar has already resigned

TechCrunch AI 50%

The director role for the Center for AI Standards and Innovation (CAISI) has become a revolving door since David Sacks left his position as czar.

https://techcrunch.com/2026/07/20/trumps-latest-ai-czar-has-already-resigned/

2 CONF

Google is working on a new AI chip designed to make Gemini more efficient

TechCrunch AI 50%

Alphabet, Google's parent company, is reportedly working on a new chip designed to make its Gemini models run much more efficiently.

https://techcrunch.com/2026/07/20/google-is-working-on-a-new-ai-chip-designed-to-make-g...

3 CONF

AI's most important protocol is getting a little bit easier to use

TechCrunch AI 50%

Under the new system, the protocol will take a looser, "stateless" approach to session IDs on the server side, similar to how most ordinary websites already work.

https://techcrunch.com/2026/07/20/ais-most-important-protocol-is-getting-a-little-bit-e...

4 CONF

Adobe camera app's new feature will critique your photos using AI

TechCrunch AI 50%

Adobe's Project Indigo can now remove all kinds of backgrounds from photos you snap using the app.

https://techcrunch.com/2026/07/20/adobe-camera-apps-new-feature-will-critique-your-phot...

5 **CONF**

YouTube clarifies policies around AI slop and upsetting videos

TechCrunch AI **50%**

YouTube has updated its monetization policies to more clearly define the kinds of AI-generated and low-quality videos that can’t earn ad revenue.

https://techcrunch.com/2026/07/20/youtube-clarifies-policies-around-ai-slop-and-upsetti...

6 **CONF**

China’s AI models have Trump’s AI world at war with itself

MIT Tech Review **50%**

This story originally appeared in The Algorithm, our weekly newsletter on AI. To get stories like this in your inbox first, sign up here. Over the weekend, several current and former advisors to President Donald Trump on AI publicly lobbed insults at the country’s leading AI...

https://www.technologyreview.com/2026/07/20/1140675/chinas-ai-models-have-trumps-ai-wor...

7 **CONF**

AI is more likely than humans to form biases when hiring

MIT Tech Review **50%**

The next time you apply for a job, AI may screen your résumé before any human sees it. But there’s good reason to question whether AI will judge you fairly. Researchers already know that LLMs pick up human biases from their training data. New research suggests that LLMs can also...

https://www.technologyreview.com/2026/07/20/1140655/ai-biases-hiring-humans/

8 **CONF**

Here are the 30,000 songs Sony is suing Udio’s AI music generator over

The Verge AI **50%**

Sony Music Entertainment has filed another lawsuit against Udio, accusing the AI music generator of infringing the copyright of more than 30,000 of its songs, ranging from Elvis Presley's Hound Dog to Beyoncé's Say My Name, and Harry Styles' As It Was. The lawsuit, filed in a...

https://www.theverge.com/tech/968375/sony-udio-lawsuit-songs-ai-copyright

9 **CONF**

SpaceX in your index fund, explained

The Verge AI **50%**

Index funds are touted as one of the safest ways to invest. Rather than picking and choosing individual stocks, index funds let you bet on the market as a whole. So what happens when a company like SpaceX - a giant gamble, and, in my opinion, terribly overpriced - is...

https://www.theverge.com/business/968257/spacex-in-your-index-fund-explained

0 **CONF**

Adobe’s ‘natural look’ camera app embraces generative AI

The Verge AI **50%**

Adobe's experimental camera app has taken an unexpected turn. After Project Indigo was launched last year to provide a "more natural (SLR-like) look" for iPhone photography, the Indigo camera app is now being updated with a suite of generative AI tools. And the change doesn't...

https://www.theverge.com/tech/967791/adobe-indigo-camera-app-ai-playground-update

1 **CONF**

China delivers a one-two punch to America’s AI dominance

The Verge AI **50%**

China's leading AI companies are ramping up the pressure on Silicon Valley, as Moonshot and Alibaba unveiled models they claim can go toe-to-toe with the best from OpenAI and Anthropic at a fraction of the cost. The rapid-fire releases suggest America's lead at the AI frontier...

https://www.theverge.com/ai-artificial-intelligence/967781/chinese-ai-models-open-sourc...

2 **CONF**

Apple’s plot to crush OpenAI

The Verge AI **50%**

Apple is suing OpenAI. The complaint is readable and intense, as these things often are, though many experts seem to think many of the allegations are just the ways things are done. So what does Apple really want here, and why is it picking such a public fight with OpenAI? On...

https://www.theverge.com/podcast/967244/apple-openai-lawsuit-vergecast

3 **CONF**

Phantom Transitions in Language Model Fine-Tuning: A Density-Matrix Analysis

ArXiv CS.AI **50%**

arXiv:2606.07559v2 Announce Type: replace-cross Abstract: Fine-tuning a language model often fails silently when its correct completion must outrank a near-synonym competitor. Cross-entropy loss falls monotonically while the correct token never overtakes the competitor. We study...

https://arxiv.org/abs/2606.07559

4 **CONF**

PlanFlip: Attacking Multi-Agent LLM Systems via Planning-Phase Prompt Injection

ArXiv CS.AI **50%**

arXiv:2607.16199v1 Announce Type: new Abstract: Multi-agent LLM systems increasingly rely on a Planner to decompose goals into sub-task sequences that downstream Executor and Critic agents execute and audit. We identify the planning phase as a critical attack surface: a single...

https://arxiv.org/abs/2607.16199

5 **CONF**

TopoTuner: Topological Finetuning of Large Language Models

ArXiv CS.AI **50%**

arXiv:2607.16637v1 Announce Type: new Abstract: Full fine-tuning remains a strong way to adapt pretrained LLMs, but it updates all weights and can be expensive. LoRA reduces the number of trainable parameters, but it does not directly answer which pretrained components should be...

https://arxiv.org/abs/2607.16637

6 **CONF**

Democratizing AI with Small Language Models: Structured Benchmarking and Parameter-Efficient Fine-Tuning for Local Deployment

ArXiv CS.AI **50%**

arXiv:2607.16202v1 Announce Type: new Abstract: AI democratization is not primarily a question of matching frontier-scale generality; it is a question of whether capable models can be selected, audited, and specialized under hardware and governance constraints that ordinary...

https://arxiv.org/abs/2607.16202

7 **CONF**

Masked Diffusion Language Models are Strong and Steerable Text-Based World Models for Agentic RL

ArXiv CS.AI **50%**

arXiv:2607.16204v1 Announce Type: new Abstract: Recent growth in reinforcement learning (RL) has surfaced a need for diverse, specialized training environments. Hand-curated environments with fixed task and reward difficulties become ineffective signals as model performance...

https://arxiv.org/abs/2607.16204

8 CONF

Automated Reinforcement Learning: An Overview

ArXiv CS.AI 50%

arXiv:2201.05000v3 Announce Type: replace-cross Abstract: Reinforcement Learning and, recently, Deep Reinforcement Learning are popular methods for solving sequential decision-making problems modeled as Markov Decision Processes. RL modeling of a problem and selecting algorithms...

https://arxiv.org/abs/2201.05000

9 CONF

Trace-Based On-Policy Distillation for Masked Diffusion Language Models

ArXiv CS.AI 50%

arXiv:2607.16872v1 Announce Type: cross Abstract: Diffusion large language models (dLLMs) are a promising alternative to autoregressive generation. However, reasoning-oriented post-training for dLLMs remains challenging. Supervised fine-tuning (SFT) for dLLMs requires dense but...

https://arxiv.org/abs/2607.16872

0 CONF

ColGraphRAG: Late-Interaction Evidence Retrieval for Multimodal GraphRAG

ArXiv CS.AI 50%

arXiv:2607.16208v1 Announce Type: new Abstract: Graph-grounded multimodal question answering organizes text, tables, and images in a structured evidence graph, yet end-to-end accuracy depends on which multimodal assets are ranked highly enough to enter downstream reasoning; for...

https://arxiv.org/abs/2607.16208

1 CONF

RAIL Guard: Closing the Evaluation-to-Remediation Gap in Responsible AI for LLM Agents

ArXiv CS.AI 50%

arXiv:2607.16215v1 Announce Type: new Abstract: Existing guardrail systems for large language model agents operate as binary classifiers that block unsafe content, leaving organizations to discard failing outputs and retry from scratch. We introduce RAIL Guard, a closed-loop...

https://arxiv.org/abs/2607.16215

2 CONF

Generalist AI Control: Towards Multi-purpose Adaptive Algorithms

ArXiv CS.AI 50%

arXiv:2607.16313v1 Announce Type: new Abstract: Traditional controllers are designed for specific systems and do not transfer across different system orders and dynamics. We present a Generalist Controller, a learning-based controller capable of controlling systems of varying...

https://arxiv.org/abs/2607.16313

3 CONF

Interactive Task Alignment as a POMDP

ArXiv CS.AI 50%

arXiv:2607.16412v1 Announce Type: new Abstract: Current benchmarks for language models primarily evaluate execution on fully specified tasks. However, real user tasks are often ambiguous. Users arrive with incomplete, exploratory, or even inconsistent goals, requiring the...

https://arxiv.org/abs/2607.16412

4 CONF

Nonuniformity Principle in Human-AI Coworking

ArXiv CS.AI 50%

arXiv:2607.16530v1 Announce Type: new Abstract: As generative AI is increasingly applied to automate multi-step and high-stake workflows, human judgment and involvement remain essential for ensuring the quality of AI-generated outputs. In practice, while it is desirable for...

https://arxiv.org/abs/2607.16530

5 CONF

From Modalities to Propositions: A Language-Centric Framework for Multimodal Intelligence

ArXiv CS.AI 50%

arXiv:2607.16560v1 Announce Type: new Abstract: We propose a language representation for multimodal data in which any observation, whether image, video, or text, is expressed as a bag of atomic propositions, simple statements about the entities, actions, and relations in a...

https://arxiv.org/abs/2607.16560

6 CONF

FST.ai 2.5: Explainable and Uncertainty-Aware AI for Olympic and Para-Taekwondo Decision Support, Athlete Digital Twins, and Federation-Scale Analytics

ArXiv CS.AI 50%

arXiv:2607.16597v1 Announce Type: new Abstract: The rapid digitalisation of elite sport has created new opportunities for integrating artificial intelligence (AI), performance analytics, and decision-support systems into athlete development and competition management. However,...

https://arxiv.org/abs/2607.16597

7 CONF

A Research Prototype for Closed-Loop Generative Design of Customized Foot Orthoses via Semantic-Physics Alignment

ArXiv CS.AI 50%

arXiv:2607.16631v1 Announce Type: new Abstract: Translating unstructured clinical prescriptions into patient-specific foot orthoses (FOs) is hindered by a semantic-physical misalignment: high-level clinical intent is not mapped deterministically onto the 3D geometric parameters...

https://arxiv.org/abs/2607.16631

8 CONF

Decision Making Needs Uncertainty Quantification [Lecture Notes]

ArXiv CS.AI 50%

arXiv:2607.14407v2 Announce Type: replace-cross Abstract: Many signal processing systems ultimately exist to {act}. Whenever the state variable that determines the action to be taken by a decision maker, or agent, is uncertain, the way that uncertainty is represented decides how...

https://arxiv.org/abs/2607.14407

9 CONF

DS@GT ARC at eRisk 2026: Hybrid Multi-Agent LLM System with Structured Algorithmic Guidance for Conversational Depression Screening

ArXiv CS.AI 50%

arXiv:2607.16712v1 Announce Type: new Abstract: We describe DS@GT's submission to the eRisk 2026 Task 1 challenge on conversational depression screening, in which systems interview LLM personas that simulate individuals with varying depression profiles and produce a Beck...

https://arxiv.org/abs/2607.16712

0 CONF

CARV: A Diagnostic Benchmark for Compositional Analogical Reasoning in Multimodal LLMs

ArXiv CS.AI 50%

arXiv:2603.27958v2 Announce Type: replace Abstract: Analogical reasoning tests a fundamental aspect of human cognition: mapping the relation from one pair of objects to another. Existing evaluations of this ability in multimodal large language models (MLLMs) overlook the ability...

https://arxiv.org/abs/2603.27958

1 **CONF**

Constrained Path Reasoning: Measuring When Committed Stages Earn Their Cost

ArXiv CS.AI 50%

arXiv:2607.17240v1 Announce Type: new Abstract: When does a committed intermediate stage in an LLM reasoning pipeline earn its cost? Constrained Path Reasoning (CPR) pairs a source-aware path hypothesis with stage-level accounting. Search generates provisional states; trusted or...

https://arxiv.org/abs/2607.17240

2 **CONF**

Supporting Autonomous Process Execution within a Multi-Perspective Constraint Frame via Numeric Planning

ArXiv CS.AI 50%

arXiv:2607.16738v1 Announce Type: new Abstract: AI-Augmented Business Process Management Systems (ABPMS) enhance traditional BPMS by leveraging advanced AI techniques to define, execute, and monitor complex process structures. Within this landscape, Framed Autonomy denotes the...

https://arxiv.org/abs/2607.16738

3 **CONF**

From Overload to Insights: How AI Agents Can Support Scientists in Analyzing Complex Data

ArXiv CS.AI 50%

arXiv:2607.16845v1 Announce Type: new Abstract: Scientists at European XFEL conduct experiments that generate very large and complex datasets. The subsequent data analysis is challenging as scientists must combine their domain expertise with facility- and software-specific...

https://arxiv.org/abs/2607.16845

4 **CONF**

AgentBrew: Lifelong Knowledge Brewing from Strong Teachers to Weak LLM Agents

ArXiv CS.AI 50%

arXiv:2607.16851v1 Announce Type: new Abstract: Deploying LLM agents typically requires a compact test-time student, even if a stronger teacher is available during training. We study knowledge brewing: distilling a teacher's interactive experience into a persistent external...

https://arxiv.org/abs/2607.16851

5 **CONF**

STAC: When Innocent Tools Form Dangerous Chains for LLM Agents

ArXiv CS.AI 50%

arXiv:2509.25624v3 Announce Type: replace-cross Abstract: As LLMs advance into autonomous agents with tool-use capabilities, they introduce security challenges that extend beyond traditional content-based LLM safety concerns. This paper introduces Sequential Tool Attack Chaining...

https://arxiv.org/abs/2509.25624

6 **CONF**

Reward-Driven LLM Agent Workflows: Synthesizing POMDP Routing and Self-Correction for Autonomous Decision-Making

ArXiv CS.AI 50%

arXiv:2607.17038v1 Announce Type: new Abstract: This paper addresses key technical challenges in current large language model (LLM) agent applications, including long-horizon planning, sparse reward attribution, and dynamic environmental interaction, by designing and optimizing...

https://arxiv.org/abs/2607.17038

7 **CONF**

When LLMs Over-Answer: Measuring and Mitigating Quality Issues in LLM-Based Hardware Description Language Question Answering

ArXiv CS.AI 50%

arXiv:2607.17063v1 Announce Type: new Abstract: The rapid advancement of large language models (LLMs) has led practitioners to increasingly rely on them for answering questions about hardware description languages (HDLs). Because HDL is ultimately synthesized into physical...

https://arxiv.org/abs/2607.17063

8 **CONF**

Is Your Model Thinking or Just Stagnating? PUMA: Diagnosing Reasoning Pathology via Phase-Momentum Alignment

ArXiv CS.AI 50%

arXiv:2607.17188v1 Announce Type: new Abstract: Test-time scaling empowers Large Reasoning Models (LRMs) to tackle complex tasks via extensive Chain-of-Thought (CoT). However, this often induces the "overthinking" paradox, where redundant reasoning increases computational...

https://arxiv.org/abs/2607.17188

9 **CONF**

Toward Anthropomorphic Dialogue: A Closed-Loop Framework for Human-Like Chat Generation, Evaluation, and Preference Alignment

ArXiv CS.AI 50%

arXiv:2607.17191v1 Announce Type: new Abstract: Human-like private chat requires more than fluent response generation: a system must preserve persona, relationship, memory, bounded knowledge, medium-specific timing, and a coherent multi-turn arc. We present AnthroDial, a...

https://arxiv.org/abs/2607.17191

0 **CONF**

A Systematic Evaluation of Trajectory Data Curation for LoRA Fine-Tuning of Code Agents

ArXiv CS.AI 50%

arXiv:2607.17205v1 Announce Type: new Abstract: Supervised fine-tuning (SFT) of open-weight LLMs on expert agent trajectories has emerged as a prominent approach to building capable code agents without reliance on proprietary models. A central yet underexplored question is how...

https://arxiv.org/abs/2607.17205

1 **CONF**

An Explicit World Model Based on Data-First Ontology: DaoQL Multimodal Storage Validation and Counterfactual Reasoning Evaluation

ArXiv CS.AI 50%

arXiv:2607.17269v1 Announce Type: new Abstract: Large language models encode world models implicitly in neural weights, which exposes four structural risks in high-precision domains such as medicine and finance: hallucination, frozen knowledge, poor explainability, and poor...

https://arxiv.org/abs/2607.17269

2 **CONF**

Learning-Driven Adaptive Audit Scheduling: A Sequential Decision Approach to Off-Chain Data Integrity

ArXiv CS.AI 50%

arXiv:2607.17305v1 Announce Type: new Abstract: We model cryptographic auditing of off-chain data as a Constrained MDP (CMDP) under partial observability: the storage node's hidden type and corruption state make the problem a POMDP, while a miss-rate ceiling rho imposes an...

https://arxiv.org/abs/2607.17305

3 **CONF**

Generalize and Guide: Decomposing Rewards for Few-Shot Inverse Reinforcement Learning

ArXiv CS.AI 50%

arXiv:2607.17760v1 Announce Type: cross Abstract: Inverse reinforcement learning (IRL) provides a powerful framework for learning from demonstrations. However, real-world tasks often exhibit substantial natural variations (e.g., picking up mugs with varying shapes), making it...

https://arxiv.org/abs/2607.17760

4 **CONF**

Self-Modifying Lean Proof Agents with Verifier-Grounded Benchmark Coevolution

ArXiv CS.AI 50%

arXiv:2607.17352v1 Announce Type: new Abstract: Designing effective Lean proof agents is a central challenge in formal mathematical reasoning. Beyond building stronger provers, recent work emphasizes the workflow around Lean: how an agent decomposes proof obligations, uses tools...

https://arxiv.org/abs/2607.17352

5 **CONF**

Quantifying Diversity of Thought: A Predictive Law of Weighted LLM Ensemble Lift

ArXiv CS.AI 50%

arXiv:2607.17384v1 Announce Type: new Abstract: This paper provides an experimentally verified formal law for calculating the uplift that diversity of thought provides in Large Language Model (LLM) ensembles. From first principles, we derive an exact decomposition of LLM...

https://arxiv.org/abs/2607.17384

6 **CONF**

Empirical Grounding Improves the Realism of LLM Agents Simulating Human Behavior During Disruptions

ArXiv CS.AI 50%

arXiv:2607.17437v1 Announce Type: new Abstract: Large language model (LLM) agents offer a generative approach to simulating human behavior under conditions that may have few or no direct historical analogues, a common challenge in disaster and infrastructure-disruption planning....

https://arxiv.org/abs/2607.17437

7 **CONF**

AEC-DS: Adaptive Erasure Coding with PDP-Triggered Reputation and QoS-Aware Migration for Decentralized Storage

ArXiv CS.AI 50%

arXiv:2607.17460v1 Announce Type: new Abstract: In decentralized storage systems, audit results are often not used directly to guide later redundancy and shard-placement decisions, which can lead to inefficient resource allocation and delayed recovery. We propose AEC-DS, a...

https://arxiv.org/abs/2607.17460

8 **CONF**

Can AI Agents Really Complete RTL-to-GDS? Lessons from Benchmarking Tool-Interactive EDA Workflows

ArXiv CS.AI 50%

arXiv:2607.17528v1 Announce Type: new Abstract: LLM-driven agent systems have emerged as a promising paradigm for electronic design automation (EDA), demonstrating strong potential for automating complex design workflows. However, existing evaluations primarily examine...

https://arxiv.org/abs/2607.17528

9 CONF

The Curvature Shadow: An Apparent Failure of Maximum-Entropy Equilibrium Selection is a Removable Artifact

ArXiv CS.AI 50%

arXiv:2607.17543v1 Announce Type: new Abstract: In two-player zero-sum games whose Nash equilibria form a convex set, regularized solvers such as Regularized Nash Dynamics (R-NaD) empirically select the maximum-entropy member: the information projection (I-projection) of a...

https://arxiv.org/abs/2607.17543

0 CONF

Retain or Consolidate? Budget-Dependent Operator Selection for Language Agent Memory

ArXiv CS.AI 50%

arXiv:2607.17545v1 Announce Type: new Abstract: Language agents depend on memory across interactions. However, the limited context windows of large language models (LLMs) and their inference costs constrain how much memory can be used at once. Existing systems mainly follow two...

https://arxiv.org/abs/2607.17545

1 CONF

Why Does Feedback-Augmented Self-Distillation Fail to Improve Retrieval-Interleaved Search Agents?

ArXiv CS.AI 50%

arXiv:2607.17558v1 Announce Type: new Abstract: On-policy self-distillation (OPSD) offers a promising approach for training large language models without relying on a separate teacher model. However, its effectiveness on complex agentic tasks remains largely unexplored. In this...

https://arxiv.org/abs/2607.17558

2 CONF

ZifaMem: Structured Memory for Persona, Preference, and Emotional Continuity in AI Companions

ArXiv CS.AI 50%

arXiv:2607.17564v1 Announce Type: new Abstract: AI companions are judged not only by single-turn fluency but by whether they sustain emotional continuity: remembering who the companion is, what the user prefers, and how the relationship has felt. We present ZifaMem, a structured...

https://arxiv.org/abs/2607.17564

3 CONF

Natural Language Access to Domain-Specific Metadata: A Reusable Framework for LLM Query Generation

ArXiv CS.AI 50%

arXiv:2607.18029v1 Announce Type: cross Abstract: Researchers need to answer ad-hoc questions about the contents of domain-specific archives but often lack the expertise to write structured queries on the metadata. We show that when domain vocabulary and semantics are captured...

https://arxiv.org/abs/2607.18029

4 CONF

Verify, Repair, Repeat, or Stop? Robust Stopping for Noisy Verify-Repair Loops in LLM Agents

ArXiv CS.AI 50%

arXiv:2607.17641v1 Announce Type: new Abstract: Verify-repair loops are a standard means for large language model (LLM) agents to correct faulty plans in code generation, mathematical reasoning, and tool use. When both the verifier and the repairer are noisy, repair can damage...

https://arxiv.org/abs/2607.17641

5 **CONF**

OrientSAM: Mitigating Camera-Centric Shortcut in Multimodal Spatial Reasoning via Orientation-Aware Spatial Alignment

ArXiv CS.AI 50%

arXiv:2607.17657v1 Announce Type: new Abstract: Multimodal large language models (MLLMs) still struggle with spatial reasoning that requires perspective transformation. In particular, they often rely on camera-centric cues rather than reasoning from the reference object's...

https://arxiv.org/abs/2607.17657

6 **CONF**

Artificial Intelligence for Understanding and Managing Transportation Behavior in Sustainable Smart Cities

ArXiv CS.AI 50%

arXiv:2607.17694v1 Announce Type: new Abstract: Urban transportation systems generate heterogeneous data, yet these data do not automatically become actionable management intelligence. This chapter adopts a behavior-centered perspective on artificial intelligence (AI), treating...

https://arxiv.org/abs/2607.17694

7 **CONF**

JOR-Bench: Japanese Operations Research Benchmarks for Large Language Models

ArXiv CS.AI 50%

arXiv:2607.16777v1 Announce Type: cross Abstract: We present JOR-Bench, a collection of five Japanese-language benchmarks for evaluating the ability of large language models (LLMs) to formulate and solve operations research (OR) problems. Each benchmark is a Japanese translation...

https://arxiv.org/abs/2607.16777

8 **CONF**

LaT: LLM-as-Trainer for Multi-Task Vehicle Routing Solvers

ArXiv CS.AI 50%

arXiv:2607.17708v1 Announce Type: new Abstract: Multi-task neural solvers aim to handle multiple Vehicle Routing Problem (VRP) variants within a unified model, avoiding separate training for each constraint combination. However, VRP variants differ in optimization difficulty,...

https://arxiv.org/abs/2607.17708

9 **CONF**

Semantically Similar, Logically Distinct: Diagnosing the Semantic-Answerability Gap in Table RAG

ArXiv CS.AI 50%

arXiv:2607.17742v1 Announce Type: new Abstract: Tables are a critical knowledge source in retrieval-augmented generation (RAG), but a retrieved table may lack sufficient evidence to answer a query, a property we call answerability. While answerability broadly concerns whether a...

https://arxiv.org/abs/2607.17742

0 **CONF**

WorldCupArena: Fine-Grained Evaluation of Language Models and Deep-Research Agents on Football Forecasting

ArXiv CS.AI 50%

arXiv:2607.18084v1 Announce Type: new Abstract: Predicting a football match before kickoff requires more than knowing past results: a model must use changing information and make a clear prediction before the answer is available. We present WorldCupArena, a dynamic benchmark for...

https://arxiv.org/abs/2607.18084

1 CONF

Dynamic Defense Profiling Enables Cognitive Jailbreak of Text-to-Image Models

ArXiv CS.AI 50%

arXiv:2607.17779v1 Announce Type: new Abstract: Text-to-Image (T2I) generative models have achieved remarkable progress in synthesizing high-quality visual content, yet they remain vulnerable to adversarial misuse, particularly in generating Not-Safe-For-Work (NSFW) images. Most...

https://arxiv.org/abs/2607.17779

2 CONF

PGN: Design and Implementation of a Vision-Language Navigation System Based on Pangu Multimodal Foundation Model

ArXiv CS.AI 50%

arXiv:2607.17806v1 Announce Type: new Abstract: Vision-Language Navigation (VLN) requires an embodied agent to interpret a natural-language instruction and predict actions from temporally ordered visual observations. Adapting a multimodal large language model to VLN requires...

https://arxiv.org/abs/2607.17806

3 CONF

A Hardware-oriented Approach for Efficient Bayesian Inference Computation and Deployment

ArXiv CS.AI 50%

arXiv:2607.17855v1 Announce Type: new Abstract: Bayesian inference provides a principled foundation for reasoning under uncertainty, but its computational cost hinders deployment on resource-constrained edge devices. In this paper, we present a hardware-oriented methodology for...

https://arxiv.org/abs/2607.17855

4 CONF

ST-Veto: Spatio-Temporal Token Veto for Diffusion MLLMs via Taylor Prediction and Visual Grounding

ArXiv CS.AI 50%

arXiv:2607.17884v1 Announce Type: new Abstract: Vision Language Models (VLMs) achieve strong reasoning with Chain-of-Thought (CoT) prompting but incur high sequential-generation cost, error accumulation, and limited self-correction. Diffusion Multimodal Large Language Models...

https://arxiv.org/abs/2607.17884

5 CONF

The Autonomous Agency Scale: A Behavioral Framework for Measuring Self-Directed Behavior in AI Systems

ArXiv CS.AI 50%

arXiv:2607.17947v1 Announce Type: new Abstract: Existing AI measurement frameworks quantify cognitive capability, task automation, or catastrophic risk, but none measure autonomous agency: the extent to which a system behaves in a self-directed way. A system can saturate...

https://arxiv.org/abs/2607.17947

6 CONF

OntoExtend: A Framework for Requirement-driven and Scalable Ontology Extension with LLMs

ArXiv CS.AI 50%

arXiv:2607.17963v1 Announce Type: new Abstract: Ontology extension refers to the process of enriching an existing ontology in response to emerging requirements, making it more complete. This task is a resource-intensive and error-prone process. Large Language Models (LLMs) have...

https://arxiv.org/abs/2607.17963

7 CONF

Do Maps Still Matter for Machines: Revisiting the Role of Choropleth Maps in Foundation Model Spatial Understanding

ArXiv CS.AI 50%

arXiv:2607.17999v1 Announce Type: new Abstract: Spatial understanding is crucial for foundation models (FMs), and maps have long helped humans organize and reason about geographic information. This study examines whether choropleth maps remain useful for machine spatial...

https://arxiv.org/abs/2607.17999

8 CONF

PAMD: Structured Adaptive Distances for Bisimulation Representations in Visual Reinforcement Learning

ArXiv CS.AI 50%

arXiv:2607.18004v1 Announce Type: new Abstract: Many visual reinforcement learning (RL) algorithms learn representations by matching latent distances to a behavioral distance induced by reward and transition similarity. In practice, the choice of the latent distance can strongly...

https://arxiv.org/abs/2607.18004

9 CONF

Rethinking Heterogeneous LLM Merging: A Weighted Model Averaging Perspective

ArXiv CS.AI 50%

arXiv:2607.18026v1 Announce Type: new Abstract: Can large language models with substantially different parameter spaces be merged by direct weighted averaging, without training or semantic alignment? Existing heterogeneous fusion methods typically introduce distillation,...

https://arxiv.org/abs/2607.18026

0 CONF

AdaHome: An Adaptive Smart Home Assistant using Local Small Language Models

ArXiv CS.AI 50%

arXiv:2607.18034v1 Announce Type: new Abstract: Smart home assistants interpret a wide range of user commands, from explicit device control to underspecified and preference dependent requests. While recent systems based on Large Language Models (LLMs) improve this capability,...

https://arxiv.org/abs/2607.18034

1 CONF

Judge-dependent safety gains and model-specific helpfulness costs of evidence-sufficiency prompting in clinical LLMs

ArXiv CS.AI 50%

arXiv:2607.18086v1 Announce Type: new Abstract: Background: LLM judges increasingly score whether clinical language models give overconfident answers under incomplete evidence, yet whether a measured "safety gain" reflects real behavior change or the judge's calibration is...

https://arxiv.org/abs/2607.18086

2 CONF

Can We Break LLMs Out of Self-Loops? Fine-Grained Reasoning Control with Activation Steering

ArXiv CS.AI 50%

arXiv:2607.18100v1 Announce Type: new Abstract: Extended reasoning has become standard for frontier Large Language Models (LLMs), yet the trajectories these models produce remain largely uncontrollable. Existing methods for shaping how a model reasons are prompt based approaches...

https://arxiv.org/abs/2607.18100

3 CONF

What Makes Linguistic Representations Good Models of High-Level Visual Perception in the Human Brain?

ArXiv CS.AI 50%

arXiv:2607.16214v1 Announce Type: cross Abstract: Image descriptions represented with language models (LMs) predict human brain responses to naturalistic images in high-level visual regions, but the factors driving this predictivity remain unclear. To investigate this, we...

https://arxiv.org/abs/2607.16214

4 CONF

Comparing Spectrogram Front-Ends for Abnormal Heart-Sound Detection with a Convolutional Neural Network

ArXiv CS.AI 50%

arXiv:2607.16220v1 Announce Type: cross Abstract: Heart disease kills a lot of people, and one cheap way to catch it early is by listening to heart sounds with a stethoscope, or better yet, just recording them and running them through a model. This project is a binary...

https://arxiv.org/abs/2607.16220

5 CONF

Fully-sensorized smart-eyewear platform for on-device Machine Learning

ArXiv CS.AI 50%

arXiv:2607.16222v1 Announce Type: cross Abstract: This paper presents ARGO, a smart eyewear platform designed to bridge ergonomic comfort, high computational throughput, and energy efficiency. Unlike cloud-dependent solutions, ARGO leverages the STM32N6 microcontroller and its...

https://arxiv.org/abs/2607.16222

6 CONF

International Agreements to Limit Frontier AI: Objectives and Exit

ArXiv CS.AI 50%

arXiv:2607.16224v1 Announce Type: cross Abstract: An international agreement to limit AI development could be crucial to mitigate risks from AI. However, it remains unclear which conditions should determine when the limiting measures are relaxed. We survey existing international...

https://arxiv.org/abs/2607.16224

7 CONF

From Weights to Words: Expressing and Editing Preference Model Inferences in Natural Language

ArXiv CS.AI 50%

arXiv:2607.16232v1 Announce Type: cross Abstract: The growing use of statistical learning algorithms to infer human preferences from high-dimensional choice data runs up against a fundamental challenge: choice alternatives typically differ in many ways simultaneously, so it is...

https://arxiv.org/abs/2607.16232

8 CONF

Token-Level Cross-Modal Transformer with Contrastive Multi-Task Learning for Breast Cancer Subtype Classification and Survival Prediction

ArXiv CS.AI 50%

arXiv:2607.16233v1 Announce Type: cross Abstract: Integrating heterogeneous genomic and clinical modalities for joint cancer subtype classification and survival prediction remains a key challenge in precision oncology. Existing approaches suffer from three limitations: (1) they...

https://arxiv.org/abs/2607.16233

9 CONF

DA-Fusion: Deformable Attention-Based RGB-D Fusion Transformer for Unseen Object Instance Segmentation

ArXiv CS.AI 50%

arXiv:2607.17754v1 Announce Type: cross Abstract: In logistics automation, precise segmentation of unseen objects is crucial for efficient robotic manipulation in cluttered environments. Tasks such as bin-picking and shelf-picking require robust perception to handle occlusions,...

https://arxiv.org/abs/2607.17754

0 CONF

Diffusion-corrected Autoregressive Fourier Neural Operator for Droplet Evolution Prediction

ArXiv CS.AI 50%

arXiv:2607.16238v1 Announce Type: cross Abstract: Predicting droplet evolution in material jetting, or Inkjet Printing (IJP), is essential for maintaining printing quality. However, long-horizon forecasts remain challenging due to error accumulation and the complex coupling of...

https://arxiv.org/abs/2607.16238

1 CONF

KernelBench-Verified: Do LLM-Generated Kernels Actually Beat PyTorch?

ArXiv CS.AI 50%

arXiv:2607.16241v1 Announce Type: cross Abstract: Recent large language models (LLMs) can generate custom CUDA kernels that appear to outperform PyTorch on benchmarks such as KernelBench. Building upon this foundational framework, we demonstrate that frontier models frequently...

https://arxiv.org/abs/2607.16241

2 CONF

RobustVLA: On Robustness of Vision-Language-Action Model against Multi-Modal Perturbations

ArXiv CS.AI 50%

arXiv:2510.00037v5 Announce Type: replace-cross Abstract: In Vision-Language-Actionf(VLA) models, robustness to real-world perturbations is critical for deployment. Existing methods target simple visual disturbances, overlooking the broader multi-modal perturbations that arise...

https://arxiv.org/abs/2510.00037

3 CONF

Learning Structural Manipulability in Gate-Level Netlists Using Graph Neural Networks

ArXiv CS.AI 50%

arXiv:2607.16245v1 Announce Type: cross Abstract: Gate-level netlists exhibit intrinsic structural properties that influence signal propagation independently of functional simulation. We define a topology-driven structural manipulability score that characterizes node-level...

https://arxiv.org/abs/2607.16245

4 CONF

Let the Data Decide: Supervision Analysis, Capability Trade-offs, and Adaptive Objective Routing in Continued Pre-Training via Off-Policy Distillation

ArXiv CS.AI 50%

arXiv:2607.16246v1 Announce Type: cross Abstract: Off-policy distillation is now central to large language model pre-training, yet how training data, objective parameterization, and model capabilities interact remains poorly characterized. We studies top-k-truncated,...

https://arxiv.org/abs/2607.16246

5 CONF

Benchmarking Machine Learning Models for Multi-Omics-Based Breast Cancer Prediction

ArXiv CS.AI 50%

arXiv:2607.16250v1 Announce Type: cross Abstract: Estrogen Receptor (ER) status is a critical biomarker in breast cancer diagnosis, prognosis, and treatment selection. Recent advances in high-throughput sequencing technologies have enabled the generation of multi-omics datasets...

https://arxiv.org/abs/2607.16250

6 CONF

Comprehensive Evaluation of Machine Learning for Type 2 Diabetes Risk Prediction: Large-Scale External Validation and Fairness Analysis

ArXiv CS.AI 50%

arXiv:2607.16253v1 Announce Type: cross Abstract: Machine learning-based Type 2 diabetes risk prediction models obtain good internal validation results but lose effectiveness in real-world applications due to deficient external testing and fairness assessment. We developed a...

https://arxiv.org/abs/2607.16253

7 CONF

Feature Generation Using LLMs: An Evolutionary Algorithm Approach

ArXiv CS.AI 50%

arXiv:2607.16255v1 Announce Type: cross Abstract: A crucial step in machine learning pipelines is to present each entity with features or attributes that are representative of the characteristics of the processed entities. Feature engineering is an important step in finding a...

https://arxiv.org/abs/2607.16255

8 CONF

Discovery by Dreaming: Cross-Domain Recombination in Artificial Memory

ArXiv CS.AI 50%

arXiv:2607.16256v1 Announce Type: cross Abstract: Dreams splice together people, places, and times that never met. Neuroscience suggests this recombination is not noise, but a function driving insight and creative discovery. This reframes memory consolidation: rather than merely...

https://arxiv.org/abs/2607.16256

9 CONF

COLIP-2: Olfaction-Vision-Language Embeddings

ArXiv CS.AI 50%

arXiv:2607.17559v1 Announce Type: cross Abstract: The Contrastive Olfaction-Language-Image Pre-training 2 (COLIP-2) model is a multimodal embeddings space that places olfaction as a first-class citizen among vision and language. Molecular structure, gas-sensor readings,...

https://arxiv.org/abs/2607.17559

00 CONF

Autonomous mechanistic discovery of colorectal cancer vulnerabilities via multi-scale AI swarms

ArXiv CS.AI 50%

arXiv:2607.16262v1 Announce Type: cross Abstract: The acceleration of automated scientific discovery has been fundamentally bottlenecked by the epistemic gap between the semantic reasoning of large language models (LLMs) and the deterministic physics of mammalian biology. While...

https://arxiv.org/abs/2607.16262

01 CONF

From Intent to Infrastructure: LLM-Driven Agent Compilers for ISAC Networks

ArXiv CS.AI 50%

arXiv:2607.16269v1 Announce Type: cross Abstract: Integrated sensing and communications (ISAC) is moving from proof-of-concept demonstrations to system-level deployment in sixth-generation (6G) networks. Because sensing and communication share hardware, spectrum, and waveform...

https://arxiv.org/abs/2607.16269

02 CONF

GenSyn10: A Multi-Generative AI Dataset For Benchmarking Image Classification

ArXiv CS.AI 50%

arXiv:2607.16283v1 Announce Type: cross Abstract: The rapid advancement of generative AI has outpaced our ability to reliably detect its outputs, particularly when detectors encounter generators they have not seen before. We introduce GenSyn10, a CIFAR-10-aligned synthetic image...

https://arxiv.org/abs/2607.16283

03 CONF

It Depends on the Dataset: When a Brain-Encoding Model's Predicted Responses Beat Their Visual Backbone for Video Memorability

ArXiv CS.AI 50%

arXiv:2607.16292v1 Announce Type: cross Abstract: Brain-encoding foundation models predict fMRI responses to video, audio, and text well enough to win the Algonauts 2025 challenge. We ask whether their predicted responses, obtained with no scanner, are a useful feature lens for...

https://arxiv.org/abs/2607.16292

04 CONF

RADD: Retrieval-Augmented Discrete Diffusion for Multi-Modal Knowledge Graph Completion

ArXiv CS.AI 50%

arXiv:2604.25693v2 Announce Type: replace Abstract: Most multi-modal knowledge graph completion (MMKGC) models use one embedding scorer to conduct both retrieval over the full entity set and final link prediction. We argue that this coupling is a core bottleneck: global...

https://arxiv.org/abs/2604.25693

05 CONF

Efficient EEG Seizure Detection Using INT8 Quantization, Channel Pruning, and Spiking Neural Networks

ArXiv CS.AI 50%

arXiv:2607.16296v1 Announce Type: cross Abstract: Continuous EEG monitoring for epilepsy is constrained by the limited power and memory budgets of wearable and implantable devices. Deep neural networks can detect seizures with high accuracy, but their computational cost and...

https://arxiv.org/abs/2607.16296

06 CONF

DAUPNet: Domain-Aware Uncertainty Modeling for Reliable Prototype Discrimination in Cross-Domain Few-Shot Semantic Segmentation

ArXiv CS.AI 50%

arXiv:2607.16308v1 Announce Type: cross Abstract: Cross-domain few-shot semantic segmentation (CD-FSS) has predominantly been formulated as learning domain-invariant representations or improving support-query correspondence. Nevertheless, large domain shifts still make prototype...

https://arxiv.org/abs/2607.16308

07 **CONF**

Monte Carlo Dropout Uncertainty and Entropy-Thresholded Selective Prediction for Architecture-Agnostic Brain Tumor MRI Triage

ArXiv CS.AI 50%

arXiv:2607.16317v1 Announce Type: cross Abstract: Deep networks now subtype brain tumors on MRI about as well as specialist readers, yet accuracy is not what keeps them out of the clinic. What matters at the point of care is whether a model's confidence can be trusted to flag...

https://arxiv.org/abs/2607.16317

08 **CONF**

Explainable Lightweight Compact Deep Models for Speech Emotion Recognition

ArXiv CS.AI 50%

arXiv:2607.16803v1 Announce Type: cross Abstract: Speech Emotion Recognition (SER) is an important component in a wide range of human-centered applications, including healthcare, customer service, and human-omputer interaction. In medical and decision-support settings, there is...

https://arxiv.org/abs/2607.16803

09 **CONF**

Privacy-Aware Synthetic Video Benchmarking and Relational Evaluation for Worker-Under-Suspended-Load Detection

ArXiv CS.AI 50%

arXiv:2607.16351v1 Announce Type: cross Abstract: Publicly shareable construction-video benchmarks remain scarce, especially for safety-critical hazards that are rare, dangerous to stage, and difficult to release. We study worker under suspended load, a relational hazard that...

https://arxiv.org/abs/2607.16351

10 **CONF**

Unsupervised Multimodal Clustering for Semantics Discovery in Multimodal Utterances

ArXiv CS.AI 50%

arXiv:2405.12775v2 Announce Type: replace-cross Abstract: Discovering the semantics of multimodal utterances is essential for understanding human language and enhancing human-machine interactions. Existing methods manifest limitations in leveraging nonverbal information for...

https://arxiv.org/abs/2405.12775

11 **CONF**

Automated Hardware Validation Test Plan Generation for Large Scale AI Datacenter Platforms Using a Generative AI Multi-Agents Architecture

ArXiv CS.AI 50%

arXiv:2607.16388v1 Announce Type: cross Abstract: Large-scale AI datacenter platforms comprise thousands of heterogeneous hardware components whose validation requires comprehensive fault injection test plans. Today these plans are authored manually: engineers review hardware...

https://arxiv.org/abs/2607.16388

12 **CONF**

Think, Plan, Paint: Layout-Aware Reasoning for Controllable Image Generation in Unified Models

ArXiv CS.AI 50%

arXiv:2607.16409v1 Announce Type: cross Abstract: Unified Multimodal Large Language Models (MLLMs) offer a promising paradigm for unifying visual understanding and generation, yet they still struggle to follow complex spatial instructions and logical constraints in controllable...

https://arxiv.org/abs/2607.16409

13 **CONF**

Signal-based Model Access Risk Analysis for AI System Operations Security

ArXiv CS.AI **50%**

arXiv:2607.16414v1 Announce Type: cross Abstract: Artificial intelligence (AI) systems are now ubiquitous across domains such as security, finance, healthcare, consumer technology, and large-scale cloud services, where they process massive volumes of data and make consequential...

https://arxiv.org/abs/2607.16414

14 **CONF**

Retrieval is Enough: Training-Free Interpretability with a Tool-Using Agent

ArXiv CS.AI **50%**

arXiv:2607.16448v1 Announce Type: cross Abstract: Interpretability methods for neural network activations span a wide cost spectrum, from cheap, training-free techniques (such as linear probes, PCA, SVD) to more expensive training-based ones (such as SAEs and activation...

https://arxiv.org/abs/2607.16448

15 **CONF**

Committed Before Reasoning: Behavioral Reproduction and Preliminary Activation-Level Evidence of Answer Pre-Commitment in an Open-Weight LLM

ArXiv CS.AI **50%**

arXiv:2607.16451v1 Announce Type: cross Abstract: Chat models sometimes commit to an answer and then produce reasoning that justifies it rather than deriving it -- even when the answer contradicts a task premise. We study a minimal probe: "I want to wash my car. The car wash is...

https://arxiv.org/abs/2607.16451

16 **CONF**

K-IPO: Kendall-constrained Importance Preserving Oversampling for Imbalanced Tabular Data

ArXiv CS.AI **50%**

arXiv:2607.16478v1 Announce Type: cross Abstract: Oversampling is widely used to address class imbalance in tabular classification, but existing methods can distort the feature importance ranking underlying model explanations. Although recent studies have quantified this...

https://arxiv.org/abs/2607.16478

17 **CONF**

Building2Building: A Large Scale Benchmark for Generalizable Real-World Reinforcement Learning

ArXiv CS.AI **50%**

arXiv:2607.16534v1 Announce Type: cross Abstract: Reinforcement learning (RL) has achieved strong results in control, yet learned policies remain brittle to changes in dynamics, action spaces, observation spaces, or goals, a critical limitation for real-world deployment....

https://arxiv.org/abs/2607.16534

18 **CONF**

Mitigating Compiler Fusion-Induced Power Bursts in Mobile NPU Inference as the Battery Depletes

ArXiv CS.AI **50%**

arXiv:2607.16555v1 Announce Type: cross Abstract: Mobile devices increasingly rely on real-time NPU inference for camera and perception workloads. Under low-voltage conditions, however, a single inference can induce an instantaneous voltage droop in the power-delivery network,...

https://arxiv.org/abs/2607.16555

19 CONF

Learning from World Feedback: Why Model Uncertainty Fails as a Risk Signal in Model-Based RL

ArXiv CS.AI 50%

arXiv:2607.16591v1 Announce Type: cross Abstract: The RLxF programme argues that learning signals should come from world feedback rather than from internal model proxies. We instantiate this position in safe model-based control and distil it into three concrete design...

https://arxiv.org/abs/2607.16591

20 CONF

DataFlow-Harness: A Grounded Code-Agent Platform for Constructing Editable LLM Data Pipelines

ArXiv CS.AI 50%

arXiv:2607.16617v1 Announce Type: cross Abstract: Large language models (LLMs) are increasingly used to automate data-processing workflows, yet coding agents typically produce scripts that are not automatically materialized as persistent, editable platform artifacts. We call...

https://arxiv.org/abs/2607.16617

21 CONF

Privacy Cost as Equity Input: A Group Fairness Criterion for Differentially Private Machine Learning

ArXiv CS.AI 50%

arXiv:2607.16620v1 Announce Type: cross Abstract: Differential privacy (DP) is increasingly deployed to limit membership inference risk in machine-learning systems. Prior work has shown that DP-SGD can widen accuracy disparities across demographic groups, but this framing treats...

https://arxiv.org/abs/2607.16620

22 CONF

TellTale: Blending Multi-Instance LoRA Text Encoders and a Zero-Shot LLM Judge for Ambivalence/Hesitancy Recognition in Videos

ArXiv CS.AI 50%

arXiv:2607.16635v1 Announce Type: cross Abstract: We present TellTale, a text-only approach to ambivalence/hesitancy (A/H) recognition in interview videos, evaluated on the BAH dataset as part of the 3rd A/H Video Recognition Challenge (11th ABAW Workshop, ECCV 2026). Although...

https://arxiv.org/abs/2607.16635

23 CONF

Falsification-Based Verification of LLM-Generated Optimization Models: Sound Test Batteries and Their Detection Limits

ArXiv CS.AI 50%

arXiv:2607.16646v1 Announce Type: cross Abstract: Large language models now translate natural-language descriptions of decision problems into solver-ready optimization models, but they fail silently. A generated model often runs and still formulates the wrong problem. This paper...

https://arxiv.org/abs/2607.16646

24 CONF

How Do You Choose Your AI Component? An Interview Study of Secure AI Integration in Practice

ArXiv CS.AI 50%

arXiv:2607.16660v1 Announce Type: cross Abstract: The increasing adoption of Large Language Models (LLMs) as AI components in modern software systems introduces distinct security risks to the software supply chain. While many considerations and safety mechanisms are in place for...

https://arxiv.org/abs/2607.16660

25 CONF

RealDESED: A Real-World Domestic Sound Event Detection Benchmark

ArXiv CS.AI 50%

arXiv:2607.16736v1 Announce Type: cross Abstract: This paper presents RealDESED, a real-world domestic sound event detection (SED) benchmark comprising 5,710 audio recordings collected by 652 participants in their homes. Each recording is between 15 and 35 seconds long and...

https://arxiv.org/abs/2607.16736

26 CONF

First-Order Predictable but Pairwise Fragile: Local Task Adaptation in Trained Transformers

ArXiv CS.AI 50%

arXiv:2607.16821v1 Announce Type: cross Abstract: Task arithmetic, sequential fine-tuning, activation steering, and first-order random search all operate through relatively small perturbations around an already trained checkpoint, and they rely on different local approximations:...

https://arxiv.org/abs/2607.16821

27 CONF

Do Speech Tokens Leak Voiceprints? Speaker Inversion Attacks Against End-to-End Speech Language Models

ArXiv CS.AI 50%

arXiv:2607.16870v1 Announce Type: cross Abstract: End-to-end speech language models increasingly represent user speech with speech tokens rather than relying exclusively on cascaded ASR--LLM--TTS pipelines. Although these tokens support expressive and low-latency spoken...

https://arxiv.org/abs/2607.16870

28 CONF

Certified-Gap Dual-Price Policies for Real-Time Truckload Bid Acceptance with Relocating, Clock-Constrained Resources

ArXiv CS.AI 50%

arXiv:2607.16891v1 Announce Type: cross Abstract: A truckload carrier must accept or reject each load tender within seconds. The decision depends on fleet state, hours-of-service (HOS) clocks, and appointment windows. We model this as a weakly coupled dynamic program in which...

https://arxiv.org/abs/2607.16891

29 CONF

A Method for Learning Value Systems in Generative AI

ArXiv CS.AI 50%

arXiv:2607.16903v1 Announce Type: cross Abstract: Value-aware AI systems require explicit computational representations of human values (groundings) and their aggregation into value systems in order to align their decisions with ours. As such representations are difficult to...

https://arxiv.org/abs/2607.16903

30 CONF

PREFAIL: Identifying Precursors to Failures in Robotic Lift-and-Place Tasks to Improve Task Execution Performance

ArXiv CS.AI 50%

arXiv:2607.16921v1 Announce Type: cross Abstract: Non-prehensile manipulation enables flexible material handling with part carriers, but friction-based support makes high-speed motions failure-prone, while slower operation increases cycle time. Proactive failure prediction is...

https://arxiv.org/abs/2607.16921

31 CONF

Pediatric Bone Age Prediction Using Deep Learning

ArXiv CS.AI 50%

arXiv:2607.16936v1 Announce Type: cross Abstract: Pediatric bone age prediction is a crucial task in clinical practice that can help diagnose endocrine disorders and provide insight into a child's growth and development. However, conventional bone age prediction methods are...

https://arxiv.org/abs/2607.16936

32 CONF

Investigation of Polycystic Ovary Syndrome (PCOS) Diagnosis Using Machine Learning Approaches

ArXiv CS.AI 50%

arXiv:2607.16941v1 Announce Type: cross Abstract: Polycystic Ovarian Syndrome (PCOS) is a widespread hormone problem for women of childbearing age. Women with PCOS may not ovulate; they might have high levels of androgens and have many small cysts on the ovaries. It can cause...

https://arxiv.org/abs/2607.16941

33 CONF

CADENCE: Closing the Reasoning Gap via Coverage-Adaptive On-Policy Distillation

ArXiv CS.AI 50%

arXiv:2607.16955v1 Announce Type: cross Abstract: On-policy knowledge distillation transfers reasoning from large teachers to compact students, but existing approaches suffer three compounding failure modes: (i) cold-start collapse, where a fresh student assigns near-zero mass...

https://arxiv.org/abs/2607.16955

34 CONF

TurboVec: A Case Study in Cost-Efficient Private Retrieval for Enterprise RAG via Codebook-Oblivious Quantization

ArXiv CS.AI 50%

arXiv:2607.16973v1 Announce Type: cross Abstract: Retrieval-Augmented Generation (RAG) systems increasingly power enterprise LLM applications, yet the vector retrieval layer introduces two underexplored challenges: (1) trained codebook quantizers may expose corpus statistics...

https://arxiv.org/abs/2607.16973

35 CONF

Alignment of a Total Automation Economy

ArXiv CS.AI 50%

arXiv:2607.17015v1 Announce Type: cross Abstract: We consider economic theory from the perspective of a total automation economy, one with no human involvement in production either in manufacturing or in management. One can naturally ask whether a total automation economy is...

https://arxiv.org/abs/2607.17015

36 CONF

Where Does Agent Reliability Come From? A Cross-Benchmark Decomposition of Verification Loops, Specialist Models, and Scaffolding in a Production Enterprise Agent

ArXiv CS.AI 50%

arXiv:2607.17044v1 Announce Type: cross Abstract: Multi-step enterprise agent tasks fail in a characteristic way: single-pass inference has no checkpoint between deciding an answer and committing to it. We study one production system (Leni) whose architecture installs such...

https://arxiv.org/abs/2607.17044

37 **CONF**

Solver-Hard Is Not Model-Hard: A Hardness-Controlled Diagnostic for LLM Constraint Reasoning

ArXiv CS.AI **50%**

arXiv:2607.17047v1 Announce Type: cross Abstract: LLM constraint reasoners are often evaluated near the random-SAT phase transition, confounding density and solver hardness. We test instance-level transfer while near-matching clause density. At aligned size bins, with...

https://arxiv.org/abs/2607.17047

38 **CONF**

ThAME: 3D Memory-Enabled Heterogeneous Accelerator for LLM Mixture of Experts

ArXiv CS.AI **50%**

arXiv:2607.17074v1 Announce Type: cross Abstract: Mixture of Experts (MoE) architectures have emerged as a dominant paradigm for scaling Large Language Models (LLMs). However, MoE inference on conventional hardware is constrained by three fundamental bottlenecks. These encompass...

https://arxiv.org/abs/2607.17074

39 **CONF**

Auto Research for Materials: Auditable AI-Scientist Workflows with Held-Out Transfer

ArXiv CS.AI **50%**

arXiv:2607.17100v1 Announce Type: cross Abstract: An AI research agent can improve the score it sees without finding a modelling change that works on new materials. We ask a stricter question. After repeated experiments, does the selected change survive on data that never...

https://arxiv.org/abs/2607.17100

40 **CONF**

Talaria: Session-Aware Serverless Serving of Hundred-Billion-Parameter LLMs

ArXiv CS.AI **50%**

arXiv:2607.17181v1 Announce Type: cross Abstract: Serverless multi-model LLM systems multiplex popularity-skewed model catalogs over shared GPU pools, yet typically schedule each request independently. Tool-using agents break this abstraction: a session repeatedly calls an LLM...

https://arxiv.org/abs/2607.17181

41 **CONF**

Auditing Question-Order Effects in Large Language Models with the QQ Equality: Mechanism Characterization and a Saturation Caveat

ArXiv CS.AI **50%**

arXiv:2607.17219v1 Announce Type: cross Abstract: Human survey respondents exhibit question-order effects that satisfy the QQ (quantum question) equality, an a priori, parameter-free prediction of the projective quantum question-order model. We develop the QQ equality into an...

https://arxiv.org/abs/2607.17219

42 **CONF**

Specifying the Delegated-Autonomy Boundary: Requirements Engineering for Agentic AI

ArXiv CS.AI **50%**

arXiv:2607.17225v1 Announce Type: cross Abstract: Agentic AI systems do not just predict or recommend; they plan, maintain state, and act in external environments with varying degrees of autonomy. This changes the requirements engineering problem in a specific and...

https://arxiv.org/abs/2607.17225

43 CONF

A Large-Scale Measurement of AI Bill of Materials Completeness in Hugging Face Models

ArXiv CS.AI 50%

arXiv:2607.17242v1 Announce Type: cross Abstract: Pretrained machine learning (ML) models help developers build ML-intensive software systems without training models from scratch. However, model repositories often provide incomplete machine-readable documentation about model...

https://arxiv.org/abs/2607.17242

44 CONF

Asynchronous Multimodal Diffusion Policy Composition via Latency-Aware Guidance Fusion

ArXiv CS.AI 50%

arXiv:2607.17257v1 Announce Type: cross Abstract: Diffusion policies have shown strong potential for robotic imitation learning, and recent extensions incorporate additional modalities to improve manipulation performance. However, these modalities often differ not only in...

https://arxiv.org/abs/2607.17257

45 CONF

AIGB-R1: Self-Evolving Generative Auto-Bidding via Hierarchical Planner-Executor Optimization

ArXiv CS.AI 50%

arXiv:2607.17281v1 Announce Type: cross Abstract: Auto-bidding plays an essential role in online advertising, automatically adjusting bids for advertisers to optimize their commercial goals. The emerging AI-Generated Bidding (AIGB) paradigm widely adopts generative modeling to...

https://arxiv.org/abs/2607.17281

46 CONF

Lookahead Branching for Neural Network Verification

ArXiv CS.AI 50%

arXiv:2607.17290v1 Announce Type: cross Abstract: In this work, we investigate the effect of lookahead branching strategies in neural network verification. We present a general recipe to integrate lookahead into any branch-and-bound verifier and demonstrate how one of the...

https://arxiv.org/abs/2607.17290

47 CONF

The Optimization Trilemma: Efficiency, Comfort and Fairness in Decentralized Multi-agent Coordination

ArXiv CS.AI 50%

arXiv:2607.17311v1 Announce Type: cross Abstract: The problem of fair multi-agent coordination in decentralized settings is one of the most pressing challenges for building efficient collaborative systems. Resource allocation is based on optimized collective arrangements...

https://arxiv.org/abs/2607.17311

48 CONF

Mathematical Discovery in the Wild: AI-Guided Proofs in Banach Space Theory

ArXiv CS.AI 50%

arXiv:2607.17388v1 Announce Type: cross Abstract: We investigate the capacity of current language models to contribute to mathematical research. In Banach space theory, AI systems generated key ideas and proofs for five new results, which were then verified and refined by...

https://arxiv.org/abs/2607.17388

49 **CONF**

Probing the Difficulty Perception Mechanism of Large Language Models

ArXiv CS.AI **50%**

arXiv:2510.05969v3 Announce Type: replace-cross Abstract: Large language models (LLMs) are increasingly deployed on complex reasoning tasks, yet little is known about their ability to internally evaluate problem difficulty, which is an essential capability for adaptive reasoning...

https://arxiv.org/abs/2510.05969

50 **CONF**

After the Euclidean Highway: Hyperbolic Expert AI as the Next Innovation

ArXiv CS.AI **50%**

arXiv:2607.17513v1 Announce Type: cross Abstract: Expert domains are trees; the Euclidean transformer is not, diluting parent-child structure exponentially at depth. The hyperbolic turn left one question unasked: not how much of a network to curve, but where curvature may touch...

https://arxiv.org/abs/2607.17513

51 **CONF**

One-step lowest-variance selection in a Gaussian random-field model motivated by masked diffusion: Total correlation and a square root collision threshold

ArXiv CS.AI **50%**

arXiv:2607.17522v1 Announce Type: cross Abstract: Motivated by confidence-guided parallel unmasking in masked discrete diffusion, we study a single selection step in a stylized Gaussian random-field model. A locally dependent nonnegative score field represents position wise...

https://arxiv.org/abs/2607.17522

52 **CONF**

CommitLLM: A Fine-Tuned Pipeline for Git Commit Message Generation

ArXiv CS.AI **50%**

arXiv:2607.17532v1 Announce Type: cross Abstract: Developers frequently write uninformative git commit messages such as "fix" or "update stuff", degrading the value of version-control history for code review, debugging, and onboarding. We present CommitLLM, a three-stage...

https://arxiv.org/abs/2607.17532

53 **CONF**

CoCurve: Cross-Module Co-Pruning Curvature for Training-Free Structured LLM Pruning

ArXiv CS.AI **50%**

arXiv:2607.17568v1 Announce Type: cross Abstract: Structured pruning compresses large language models (LLMs) by removing whole computational units, such as attention heads and feed-forward (FFN) channel groups. Most training-free methods, however, rank these units independently,...

https://arxiv.org/abs/2607.17568

54 **CONF**

Parallel Decoder Transformer: Planner-Conditioned Latent Coordination for Model-Intrinsic Parallel Generation

ArXiv CS.AI **50%**

arXiv:2512.10054v3 Announce Type: replace Abstract: Autoregressive language models expose one causal token frontier, even when the requested document contains sections that could be developed concurrently. Existing parallel-generation systems arrange external branches around an...

https://arxiv.org/abs/2512.10054

55 CONF

Detection, Attribution, Narration: An End-to-End Pipeline for Explainable Money Mule Identification

ArXiv CS.AI 50%

arXiv:2607.17586v1 Announce Type: cross Abstract: Money mule accounts are critical facilitators of financial fraud, yet detecting them at scale remains challenging due to the heterogeneous nature of transactional and behavioural data. We present an end-to-end pipeline for...

https://arxiv.org/abs/2607.17586

56 CONF

Re-Sonance: A Dysarthric Asynchronous Real-Time Speech Conversion System Based on a Three-Stage Cascaded ASR-LLM-TTS Architecture

ArXiv CS.AI 50%

arXiv:2607.17615v1 Announce Type: cross Abstract: Individuals with dysarthria face significant challenges in professional speaking scenarios such as conferences, presentations, and meetings, where real-time communication is crucial. While existing Augmentative and Alternative...

https://arxiv.org/abs/2607.17615

57 CONF

Beyond Objective Expressivity: Geometry Preservation in Multimodal Contrastive Learning

ArXiv CS.AI 50%

arXiv:2607.17673v1 Announce Type: cross Abstract: Contrastive learning is increasingly moving toward settings with three or more modalities instead of image-text pairs. Yet, extending models from pairwise to higher-order multimodal alignment can introduce optimization and...

https://arxiv.org/abs/2607.17673

58 CONF

MXSens: Sensitivity-Aware Mixed-Precision Quantization for Efficient LLM Inference

ArXiv CS.AI 50%

arXiv:2607.17733v1 Announce Type: cross Abstract: 4-bit quantization enables efficient LLM inference, but suffers from significant accuracy degradation due to outliers. Prior work addresses this problem via data rotation or mixed-precision integer quantization, but often relies...

https://arxiv.org/abs/2607.17733

59 CONF

FIFA World Cup 2026 as a Contamination-Free Benchmark for LLM Forecasting Agents: Four Models, a Bookmaker, and 104 Matches

ArXiv CS.AI 50%

arXiv:2607.17765v1 Announce Type: cross Abstract: We introduce WC2026-Agents, a benchmark and dataset for evaluating large language models (LLMs) as autonomous forecasting agents on real, future events. For every one of the 104 matches of the 2026 FIFA World Cup, four frontier...

https://arxiv.org/abs/2607.17765

60 CONF

Persona-as-Configuration: Generative Stakeholder Reporting for Agricultural Floods

ArXiv CS.AI 50%

arXiv:2607.17774v1 Announce Type: cross Abstract: Cyber-physical systems built on deterministic edge inference, such as on-vehicle flood detection for agricultural fields, produce structured decision logs that must be interpreted differently by heterogeneous stakeholders....

https://arxiv.org/abs/2607.17774

61 CONF

BrainNext: A General-Purpose Self-Supervised Foundation Model for Brain MRI Analysis

ArXiv CS.AI 50%

arXiv:2607.17782v1 Announce Type: cross Abstract: Foundation models pretrained using self-supervised learning have transformed computer vision by learning transferable representations from large-scale unlabeled data. However, existing foundation models for neuroimaging remain...

https://arxiv.org/abs/2607.17782

62 CONF

Feature Attribution-Based Explainability Analysis of Deep Learning Models in Predictive Process Monitoring

ArXiv CS.AI 50%

arXiv:2607.17783v1 Announce Type: cross Abstract: Predictive process monitoring supports the optimization and control of operational business processes by forecasting the future state or outcome of ongoing cases. While deep neural networks have achieved strong performance for...

https://arxiv.org/abs/2607.17783

63 CONF

Medical Imaging Fusing Vision Transformer: Laryngeal Cancer Screening with Explanation

ArXiv CS.AI 50%

arXiv:2607.17789v1 Announce Type: cross Abstract: Early and timely screening of laryngeal cancer is crucial for improving clinical outcomes. In recent years, NBI endoscopy has become a standard diagnostic tool for the detection of laryngeal lesions. However, its effective use...

https://arxiv.org/abs/2607.17789

64 CONF

I wanted it to feel more personal: Customization of social AI as AI individualism in practice

ArXiv CS.AI 50%

arXiv:2607.17826v1 Announce Type: cross Abstract: Despite the growing availability of customizable social artificial intelligence (AI), such as ChatGPT, Grok, and Character.ai, we know little about how users actively shape social AI to reflect their personal preferences. This...

https://arxiv.org/abs/2607.17826

65 CONF

Zero Hallucination, by Construction: Hallucination-Aware Layered Oversight for Trustworthy Enterprise AI

ArXiv CS.AI 50%

arXiv:2607.17883v1 Announce Type: cross Abstract: Enterprises will not deploy AI agents they cannot trust, and the most-cited reason for distrust is hallucination: confident, fluent output that is simply not true. The common response is to wait for a model that does not...

https://arxiv.org/abs/2607.17883

66 CONF

The Aura in the Machine: Genealogy and the Status of the Work of Art in the Generative Era

ArXiv CS.AI 50%

arXiv:2607.17940v1 Announce Type: cross Abstract: This paper frames Generative Artificial Intelligence (AI) not as an unprecedented technological rupture, but as an industrial-scale manifestation of a deeply rooted historical process. Through a genealogy of generative arts, it...

https://arxiv.org/abs/2607.17940

67 CONF

Self-State Attacks on Self-Hosted AI Agents: How Far Can OS Defenses Go?

ArXiv CS.AI 50%

arXiv:2607.17986v1 Announce Type: cross Abstract: Self-hosted AI agents read and write their own memory and configuration files to function. An agent may get compromised via corruption of its own state -- a compromise realized via legitimate OS system call invocation. We refer...

https://arxiv.org/abs/2607.17986

68 CONF

How Does Alignment Tuning Shape Representations of Sycophancy and Related Cue-Induced Biases in LLMs?

ArXiv CS.AI 50%

arXiv:2607.18114v1 Announce Type: cross Abstract: Modern LLMs are alarmingly susceptible to surprisingly simple immaterial changes of input prompts: a casual hint, an incorrectly labeled few-shot example, or a fake prior assistant turn often flips an originally correct answer....

https://arxiv.org/abs/2607.18114

69 CONF

Do Language Models Dream of Binding Molecules? Benchmarking LLMs under Spatial Constraints

ArXiv CS.AI 50%

arXiv:2607.18144v1 Announce Type: cross Abstract: Structure-based drug design (SBDD) leverages the 3D structure of protein targets, often complemented by other spatial constraints, to generate candidate binding molecules. While diffusion models have dominated as a leading...

https://arxiv.org/abs/2607.18144

70 CONF

LLMs and Agentic AI Systems for Smart Grids: A Tutorial on Architectures and Applications

ArXiv CS.AI 50%

arXiv:2607.18147v1 Announce Type: cross Abstract: Large language models (LLMs) and agentic AI systems have evolved from natural language tasks to using external tools to plan, retrieve, and act in technical domains. In smart grids, recent work applies agentic schemes to...

https://arxiv.org/abs/2607.18147

71 CONF

GigaPath-Flash and GigaTIME-Flash: Efficient Pathology Foundation Models for Whole-Slide and Tumor Microenvironment Analysis

ArXiv CS.AI 50%

arXiv:2607.18218v1 Announce Type: cross Abstract: Foundation models have emerged as a driving force in computational pathology, with the potential to transform cancer diagnosis, prognosis, and treatment selection by learning transferable representations from large-scale...

https://arxiv.org/abs/2607.18218

72 CONF

Enhancing LLMs' Clinical Reasoning with Real-World Data from a Nationwide Sepsis Registry

ArXiv CS.AI 50%

arXiv:2505.02722v2 Announce Type: replace Abstract: Although large language models (LLMs) have demonstrated impressive reasoning capabilities across general domains, their effectiveness in real-world clinical practice remains limited. This is likely due to their insufficient...

https://arxiv.org/abs/2505.02722

73 CONF

AI sustains higher strategic tension than humans in chess

ArXiv CS.AI 50%

arXiv:2508.13213v4 Announce Type: replace Abstract: Strategic decision-making requires balancing immediate opportunities against long-term objectives: a tension fundamental to competitive environments. We investigate this trade-off in chess by analyzing the dynamics of human and...

https://arxiv.org/abs/2508.13213

74 CONF

Multi-modal cross-domain mixed fusion model with dual disentanglement for fault diagnosis under unseen working conditions

ArXiv CS.AI 50%

arXiv:2512.24679v2 Announce Type: replace Abstract: Intelligent fault diagnosis has become an indispensable technique for ensuring machinery reliability. However, existing methods suffer significant performance decline in real-world scenarios where models are tested under unseen...

https://arxiv.org/abs/2512.24679

75 CONF

SCA: Segment-Wise CoT Compression with Answer Alignment

ArXiv CS.AI 50%

arXiv:2603.07598v2 Announce Type: replace Abstract: Chain-of-thought (CoT) reasoning improves problem solving, but long think traces increase inference cost. Existing CoT compression methods usually optimize completion-level length. For structured thinking models, however, a...

https://arxiv.org/abs/2603.07598

76 CONF

Agent psychometrics: Task-level performance prediction in agentic coding benchmarks

ArXiv CS.AI 50%

arXiv:2604.00594v2 Announce Type: replace Abstract: As the focus in LLM-based coding shifts from static single-step code generation to multi-step agentic interaction with tools and environments, understanding which tasks will challenge agents and why becomes increasingly...

https://arxiv.org/abs/2604.00594

77 CONF

Information-Theoretic Measures in AI: A Practical Decision Framework

ArXiv CS.AI 50%

arXiv:2604.23716v3 Announce Type: replace Abstract: Information-theoretic (IT) measures are ubiquitous in artificial intelligence: entropy drives decision-tree splits and uncertainty quantification, cross-entropy is the default classification loss, mutual information underpins...

https://arxiv.org/abs/2604.23716

78 CONF

AI for Auto-Research: Roadmap & User Guide

ArXiv CS.AI 50%

arXiv:2605.18661v2 Announce Type: replace Abstract: AI-assisted research is crossing a threshold: fully automated systems can now generate research papers for as little as $15, while long-horizon agents can execute experiments, draft manuscripts, and simulate critique with...

https://arxiv.org/abs/2605.18661

79 CONF

Scientific reasoning does not reliably translate into scientific forecasting in frontier AI

ArXiv CS.AI 50%

arXiv:2605.22681v2 Announce Type: replace Abstract: AI systems are increasingly used to support forward-looking scientific judgment, but it remains unclear whether they can form reliable expectations about future scientific advances. Here we show that strong scientific reasoning...

https://arxiv.org/abs/2605.22681

80 CONF

CarbonNet: How Computer Vision Plays a Role in Climate Change? Application: Learning Geomechanics from Subsurface Geometry of CCS to Mitigate Global Warming

ArXiv CS.AI 50%

arXiv:2403.06025v4 Announce Type: replace-cross Abstract: We introduce a new approach using computer vision to predict the land surface displacement from subsurface geometry images for Carbon Capture and Sequestration (CCS). CCS has been proved to be a key component for a carbon...

https://arxiv.org/abs/2403.06025

81 CONF

Mediator: Memory-efficient LLM Merging with Less Parameter Conflicts and Uncertainty Based Routing

ArXiv CS.AI 50%

arXiv:2502.04411v3 Announce Type: replace-cross Abstract: Model merging aggregates Large Language Models (LLMs) finetuned on different tasks into a stronger one. However, parameter conflicts between models leads to performance degradation in averaging. While model routing...

https://arxiv.org/abs/2502.04411

82 CONF

Learning MMSE Filters for OFDM Channel Estimation: Attention Transformer Gains at Linear Inference

ArXiv CS.AI 50%

arXiv:2506.00452v5 Announce Type: replace-cross Abstract: In orthogonal frequency division multiplexing (OFDM), accurate channel estimation is crucial. Classical signal processing-based approaches, such as linear minimum mean-squared error (LMMSE) estimation, often require...

https://arxiv.org/abs/2506.00452

83 CONF

"Not in My Backyard": LLMs Uncover Online and Offline Social Biases Against Homelessness

ArXiv CS.AI 50%

arXiv:2508.13187v4 Announce Type: replace-cross Abstract: Homelessness is a persistent social challenge, impacting millions worldwide. Over 876,000 people experiencing homelessness (PEH) were recorded in the U.S. in 2025. Social bias is a significant barrier to alleviating...

https://arxiv.org/abs/2508.13187

84 CONF

Is "Knowing It's Malicious Enough?" Evaluating LLMs for Fine-Grained Malware Behavior Auditing

ArXiv CS.AI 50%

arXiv:2509.14335v2 Announce Type: replace-cross Abstract: Automated malware classifiers achieve strong detection performance, but auditing requires more than flagging a sample: analysts must explain malicious behaviors and justify them with code evidence. Traditional...

https://arxiv.org/abs/2509.14335

85 CONF

When Fewer Layers Break More Chains: Layer Pruning Harms Test-Time Scaling in LLMs

ArXiv CS.AI 50%

arXiv:2510.22228v2 Announce Type: replace-cross Abstract: Layer pruning has emerged as a widely adopted technique for improving the efficiency of large language models (LLMs). Although existing methods demonstrate strong performance retention on general knowledge tasks, their...

https://arxiv.org/abs/2510.22228

86 CONF

BBOPlace-Bench: Benchmarking Black-Box Optimization for Chip Placement

ArXiv CS.AI 50%

arXiv:2510.23472v2 Announce Type: replace-cross Abstract: Chip placement is a vital stage in modern chip design, and black-box optimization (BBO) has been applied to it for decades. Early BBO efforts, however, were limited by immature problem formulations and inefficient...

https://arxiv.org/abs/2510.23472

87 CONF

CORE -- A Cell-Level Coarse-to-Fine Image Registration Engine for Multi-stain Image Alignment

ArXiv CS.AI 50%

arXiv:2511.03826v4 Announce Type: replace-cross Abstract: Accurate and efficient registration of whole slide images (WSIs) is essential for high-resolution, nuclei-level analysis in multi-stained tissue slides. We propose a novel coarse-to-fine framework CORE for accurate...

https://arxiv.org/abs/2511.03826

88 CONF

mHC-GNN: Manifold-Constrained Hyper-Connections for Graph Neural Networks

ArXiv CS.AI 50%

arXiv:2601.02451v2 Announce Type: replace-cross Abstract: Graph Neural Networks (GNNs) suffer from over-smoothing in deep architectures and expressiveness bounded by the 1-Weisfeiler-Leman (1-WL) test. We adapt Manifold-Constrained Hyper-Connections, recently proposed for...

https://arxiv.org/abs/2601.02451

89 CONF

Lil: Less is Less When Applying Post-Training Sparse-Attention Algorithms in Long-Decode Stage

ArXiv CS.AI 50%

arXiv:2601.03043v4 Announce Type: replace-cross Abstract: Large language models (LLMs) demonstrate strong capabilities across a wide range of complex tasks and are increasingly deployed at scale, placing significant demands on inference efficiency. Prior work typically...

https://arxiv.org/abs/2601.03043

90 CONF

ReMIND: Orchestrating Modular Large Language Models for Controllable Serendipity A REM-Inspired System Design for Emergent Creative Ideation

ArXiv CS.AI 50%

arXiv:2601.07121v2 Announce Type: replace-cross Abstract: Large language models (LLMs) are increasingly used not only for problem solving but also for creative ideation; however, generating ideas that are both novel and coherent remains challenging. While high-temperature...

https://arxiv.org/abs/2601.07121

91 CONF

Evaluating LLMs When They Do Not Know the Answer: Statistical Evaluation of Mathematical Reasoning via Comparative Signals

ArXiv CS.AI 50%

arXiv:2602.03061v2 Announce Type: replace-cross Abstract: Evaluating mathematical reasoning in LLMs is constrained by limited benchmark sizes and inherent model stochasticity, yielding high-variance accuracy estimates and unstable rankings across platforms. On difficult...

https://arxiv.org/abs/2602.03061

92 CONF

UnMaskFork: Test-Time Scaling for Masked Diffusion via Deterministic Action Branching

ArXiv CS.AI 50%

arXiv:2602.04344v2 Announce Type: replace-cross Abstract: Test-time scaling strategies have effectively leveraged inference-time compute to enhance the reasoning abilities of Autoregressive Large Language Models. In this work, we demonstrate that Masked Diffusion Language Models...

https://arxiv.org/abs/2602.04344

93 CONF

NanoZK: Privacy-Preserving Verifiable Inference for Large Language Models via Layerwise Zero-Knowledge Proofs

ArXiv CS.AI 50%

arXiv:2603.18046v2 Announce Type: replace-cross Abstract: We present NanoZK, a zero-knowledge proof system for verifiable LLM inference: clients and third-party auditors check that a provider executed the advertised model on a committed input without learning weights or...

https://arxiv.org/abs/2603.18046

94 CONF

FETS Benchmark: Foundation Models Enable Scalable and Generalizable Energy Time Series Forecasting

ArXiv CS.AI 50%

arXiv:2604.22328v2 Announce Type: replace-cross Abstract: Driven by the transition towards a climate-neutral energy system, accurate energy time series forecasting is critical for planning and operations. Yet, it remains a dataset-specific task, requiring comprehensive training...

https://arxiv.org/abs/2604.22328

95 CONF

Do We Really Need Quantum Machine Learning?: A Multidimensional Empirical Study

ArXiv CS.AI 50%

arXiv:2605.27923v2 Announce Type: replace-cross Abstract: The rapid growth of computer vision and increasingly complex image recognition tasks has exposed fundamental computational limitations of classical machine learning models, motivating the exploration of quantum computing...

https://arxiv.org/abs/2605.27923

96 CONF

AI Contagion in Social Networks

ArXiv CS.AI 50%

arXiv:2606.15206v2 Announce Type: replace-cross Abstract: We study how artificial intelligence (AI) interacts with social communication networks to shape the stability of collective knowledge. Agents exchange information through a network while AI systems generate content and...

https://arxiv.org/abs/2606.15206

97 CONF

Short-Term Electricity Demand Forecasting for New England: A Comprehensive Machine Learning Benchmark with Weather, Calendar, and COVID-19 Indicators

ArXiv CS.AI 50%

arXiv:2606.20918v2 Announce Type: replace-cross Abstract: Accurate short-term electricity demand forecasting is critical for reliable power system operation, energy market planning, and infrastructure optimization. This paper benchmarks ten machine learning models for daily...

https://arxiv.org/abs/2606.20918

98 CONF

Trusting sovereign language models as scientific instruments: evidence from Portugal's AMALIA

ArXiv CS.AI 50%

arXiv:2607.08731v2 Announce Type: replace-cross Abstract: National language models are becoming publicly funded epistemic infrastructure. Public ownership, linguistic specialization, and open weights create a presumption of trustworthiness. Such an instrument, built by and for a...

https://arxiv.org/abs/2607.08731

99 CONF

A small language model detects behavioural faithfulness gaps that frontier judges and human raters miss

ArXiv CS.AI 50%

arXiv:2607.09306v2 Announce Type: replace-cross Abstract: Whether a language model behaves as it claims is a judgement on which independent human raters cannot agree (Fleiss kappa = 0.074). We show that a small, purpose-built instrument does better. A linear read-out of the...

https://arxiv.org/abs/2607.09306

00 CONF

Scalable Visual Pretraining for Language Intelligence

ArXiv CS.AI 50%

arXiv:2607.09657v2 Announce Type: replace-cross Abstract: The rapid progress of large foundation models has been driven predominantly by pretraining on large-scale text corpora. However, many forms of knowledge are conveyed through visual representations, where figures, typeset...

https://arxiv.org/abs/2607.09657